

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)****Department of Computer Science and Engineering****ASSESSMENT TOOL 2 – Industry Problem Solving**

|                         |                                                                                                                                     |                      |                                               |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------|
| <b>Course Code:</b>     | <b>DSA0404</b>                                                                                                                      | <b>Course Title:</b> | <b>FUNDAMENTALS OF DATA SCIENCE</b>           |
| <b>CO Assessed:</b>     | <b>CO4 – Industry Problem Solving</b>                                                                                               | <b>Assessment:</b>   | <b>Assessment Tool 1– Poster Presentation</b> |
| <b>Weightage:</b>       | <b>40% of CIA</b>                                                                                                                   | <b>Total Marks:</b>  | <b>100</b>                                    |
| <b>CO3 Statement:</b>   | Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4) |                      |                                               |
| <b>Student Name:</b>    | B. Ganga Jayadev Reddy                                                                                                              | <b>Reg. No.:</b>     | 192472248                                     |
| <b>Section / Batch:</b> | CSE AI                                                                                                                              | <b>Date:</b>         | 21 - 08 - 2026                                |

**Code of Conduct**

I B. Ganga Jayadev Reddy (192472248) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: B. Ganga Jayadev Reddy

| Criteria                | Weight age | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                    | Level 1 – Beginning                                  | Marks |
|-------------------------|------------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------|------------------------------------------------------|-------|
| Understanding of Case   | 25         | Thorough understanding; clearly identifies all key issues        | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |       |
| Application of Concepts | 25         | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts         | Fails to apply relevant concepts                     |       |
| Analysis                | 20         | I n - d e p t h analysis with s t r o n g reasoning and insights | Good analysis with logical reasoning           | Basic analysis with limited depth       | Weak or no analysis; lacks reasoning                 |       |
| Solution Design         | 20         | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification | Incorrect or no solution provided                    |       |
| Clarity                 | 10         | Clear, well-structured, and easy to and                          | Organized and understandable presentation      | Limited clarity and structure           | Poor clarity; difficult to understand                |       |

## 1. Marketing

**Question:** Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

**1. Identify the Population and Sample:** Population: All visitors to the marketing campaign.

Sample: 1,000 visitors.

**2. Identify the Parameter of Interest:** Population conversion rate ( $p$ ).

**3. Calculate the Point Estimate:**  $\hat{p} = 90/1000 = 0.09$  (9%)

**4. State the Null Hypothesis ( $H_0$ ):**  $H_0: p = 0.08$

**5. State the Alternative Hypothesis ( $H_1$ ):**  $H_1: p > 0.08$

**6. Select the Significance Level ( $\alpha$ ):**  $\alpha = 0.05$

**7. Calculate the Confidence Interval and/or Test Statistic:**  $SE =$

$$\sqrt{[(0.09 \times 0.91)/1000]} = 0.00905$$

$$ME = 1.96 \times 0.00905 = 0.0177$$

$$95\% \text{ CI} = 0.09 \pm 0.0177 = (0.0723, 0.1077)$$

$$95\% \text{ CI} = (7.23\%, 10.77\%)$$

**8. Calculate the P-value:**  $z = (0.09 - 0.08)/\sqrt{[(0.08 \times 0.92)/1000]} = 1.17$

$$P\text{-value} = 0.122$$

**9. Compare P-value with  $\alpha$ :** Since  $0.122 > 0.05$ , Do Not Reject  $H_0$ .

**10. Make an Industry-Oriented Conclusion:** The campaign increased the conversion rate from 8% to 9%, but the improvement is not statistically significant. More sample data is needed.

## 2. Education Technology

**Question:** An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students has an average score of 78 with a standard deviation of 10. Use a 95% confidence interval to assess the claim.

**1. Identify the Population and Sample:** Population: All students using the adaptive-learning platform.

Sample: 64 students.

**2. Identify the Parameter of Interest:** Population mean test score ( $\mu$ ).

**3. Calculate the Point Estimate:**  $\bar{x} = 78$

**4. State the Null Hypothesis ( $H_0$ ):**  $H_0: \mu = 75$

**5. State the Alternative Hypothesis ( $H_1$ ):**  $H_1: \mu > 75$

**6. Select the Significance Level ( $\alpha$ ):**  $\alpha = 0.05$

**7. Calculate the Confidence Interval and/or Test Statistic:**  $SE = 10/\sqrt{64} = 1.25$

$ME = 1.96 \times 1.25 = 2.45$

$95\% \text{ CI} = 78 \pm 2.45 = (75.55, 80.45)$

**8. Calculate the P-value:**  $z = (78 - 75)/1.25 = 2.40$

$P\text{-value} = 0.0082$

**9. Compare P-value with  $\alpha$ :** Since  $0.0082 < 0.05$ , Reject  $H_0$ .

**10. Make an Industry-Oriented Conclusion:** The adaptive-learning platform significantly improves student performance, supporting the company's claim.

### 3. Automobile Industry

**Question:** A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars has a mean mileage of 19.2 km/litre and a standard deviation of 2.5. Perform a hypothesis test using the p-value approach.

**1. Identify the Population and Sample:** Population: All cars of the new model.

Sample: 50 cars.

**2. Identify the Parameter of Interest:** Population mean mileage ( $\mu$ ).

**3. Calculate the Point Estimate:**  $\bar{x} = 19.2$  km/litre

**4. State the Null Hypothesis ( $H_0$ ):**  $H_0: \mu = 20$

**5. State the Alternative Hypothesis ( $H_1$ ):**  $H_1: \mu \neq 20$

**6. Select the Significance Level ( $\alpha$ ):**  $\alpha = 0.05$

**7. Calculate the Confidence Interval and/or Test Statistic:**  $SE = 2.5/\sqrt{50} = 0.354$

$$z = (19.2 - 20)/0.354 = -2.26$$

**8. Calculate the P-value:** P-value = 0.0236

**9. Compare P-value with  $\alpha$ :** Since  $0.0236 < 0.05$ , Reject  $H_0$ .

**10. Make an Industry-Oriented Conclusion:** The sample shows the average mileage is significantly lower than 20 km/litre. The manufacturer should review its claim.

## 4. Cloud Computing

**Question:** A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and a standard deviation of 12 ms. Test the claim using the p-value method.

**1. Identify the Population and Sample:** Population: All server requests.

Sample: 81 requests.

**2. Identify the Parameter of Interest:** Population mean response time ( $\mu$ ).

**3. Calculate the Point Estimate:**  $\bar{x} = 104$  ms

**4. State the Null Hypothesis ( $H_0$ ):**  $H_0: \mu = 100$  ms

**5. State the Alternative Hypothesis ( $H_1$ ):**  $H_1: \mu \neq 100$  ms

**6. Select the Significance Level ( $\alpha$ ):**  $\alpha = 0.05$

**7. Calculate the Confidence Interval and/or Test Statistic:**  $SE = 12/\sqrt{81} = 1.333$

$z = (104 - 100)/1.333 = 3.00$

**8. Calculate the P-value:** P-value = 0.0027

**9. Compare P-value with  $\alpha$ :** Since  $0.0027 < 0.05$ , Reject  $H_0$ .

**10. Make an Industry-Oriented Conclusion:** The average server response time is significantly higher than 100 ms. Server optimization is recommended.

## 5. E-commerce

**Question:** An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10%.

**1. Identify the Population and Sample:** Population: All customer orders.

Sample: 1,000 orders.

**2. Identify the Parameter of Interest:** Population return rate ( $p$ ).

**3. Calculate the Point Estimate:**  $\hat{p} = 130/1000 = 0.13$  (13%)

**4. State the Null Hypothesis ( $H_0$ ):**  $H_0: p = 0.10$

**5. State the Alternative Hypothesis ( $H_1$ ):**  $H_1: p > 0.10$

**6. Select the Significance Level ( $\alpha$ ):**  $\alpha = 0.05$

**7. Calculate the Confidence Interval and/or Test Statistic:**  $SE =$

$$\sqrt{[(0.10 \times 0.90)/1000]} = 0.00949$$

$$z = (0.13 - 0.10)/0.00949 = 3.16$$

**8. Calculate the P-value:** P-value = 0.0008

**9. Compare P-value with  $\alpha$ :** Since  $0.0008 < 0.05$ , Reject  $H_0$ .

**10. Make an Industry-Oriented Conclusion:** The return rate is significantly higher than 10%. The company should improve product quality and customer satisfaction.